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This apical cell governs the growth and development of whole plant body. It is applicable in Algae, Bryophytes and in some Pteridophytes.

Histogen Theory

Histogen theory is proposed by **Hanstein** (1868) and supported by **Strassburgur.** The shoot apex comprises three distinct zones.

1. **Dermatogen:** It is a outermost layer. It gives rise to epidermis.
2. **Periblem:** It is a middle layer. It gives rise to cortex.
3. **Plerome:** It is innermost layer. It gives rise to stele

Tunica Corpus Theory

Tunica corpus theory is proposed by **A. Schmidt** (1924).

Two zones of tissues are found in apical meristem.

1. **The tunica**: It is the peripheral zone of shoot apex, that forms epidermis.
2. **The corpus**: It is the inner zone of shoot apex,that forms cortex and stele of shoot.

Root Apical Meristem

Root apex is present opposite to the shoot apex. The roots contain root cap at their apices and the apical meristem is present below the root cap. The different theories proposed to explain root apical meristem organization is given below.

Apical Cell Theory

Apical cell theory is proposed by **Nageli.** The single apical cell or apical initial composes the root meristem. The apical initial is tetrahedral in shape and produces root cap from one side. The remaining three sides produce epidermis, cortex and vascular tissues. It is found in vascular cryptogams.

Histogen Theory

Histogen theory is proposed by **Hanstein (1868)** and supported by **Strassburgur**. The histogen theory as appilied to the root apical meristem speaks of four histogen in the meristem. They are respectively

i. **Dermatogen:** It is a outermost layer. It gives rise to root epidermis.

ii. **Periblem:** It is a middle layer. It gives rise to cortex**.**

iii. **Plerome:** It is innermost layer. It gives rise to stele

iv. **Calyptrogen:** It gives rise to root cap.

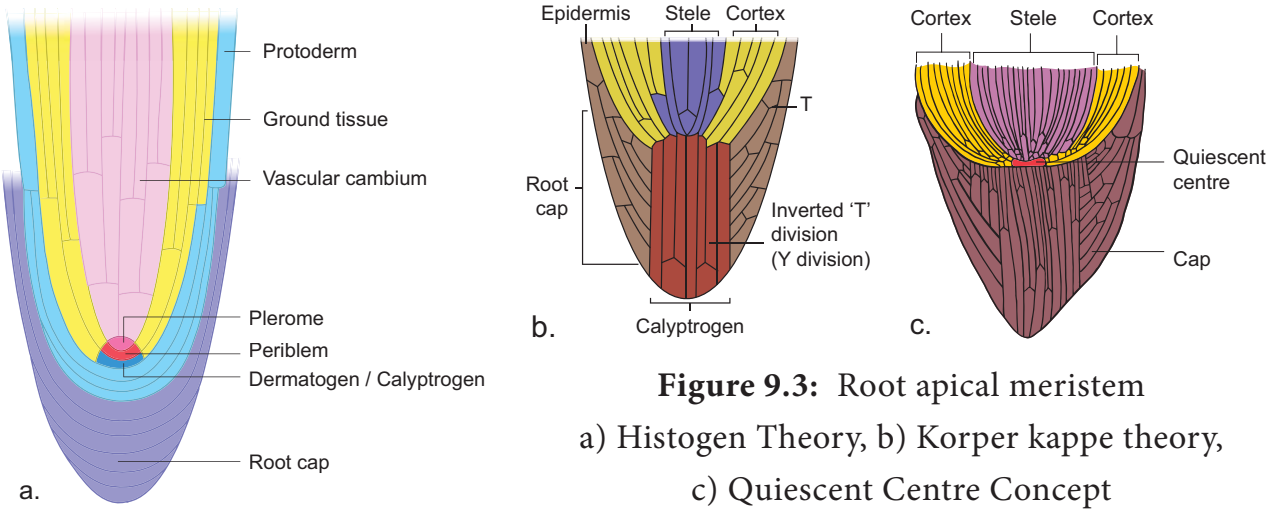


Figure 9.3: Root apical meristem
a) Histogen Theory, b) Korper kappe theory,
c) Quiescent Centre Concept